Computer Network

Experiment – Study of LAN Topology Network devices and address configuration.

Objective –

(1) To study the LAN Topology used in college campus.
(2) To identify networking device & understand their function.
(3) To analyze wired & wireless communication method used in the network.
(4) To measure speed of Network for wired & wireless connections.
(5) To obtained MAC, IP Address & subnet mask.

Problem Statement – 1

- Draw and identify LAN Topology of college campus & clearly label all components such as routers, Switches & etc.

Ans :- The college campus network observed this study consists of:

* layer – 3 core switches in the server Room.
* Managed by POE+ switches.
* Wireless Access point (APs)
* Ethernet port in classroom & Laboratories

→ Network Topology

Hybrid Hierarchical Topology which Combines:

(i) Tree Topology
(ii) Extended star Topology
(iii) Ring Topology (server room)

→ The network hierarchical layer

(i) Core layer
(ii) Distribution layer
(iii) Access layer

→ Core layer

* Core layer contains Server Room and server room have firewall, Router. 2-L3+ Switches patch Rack, Server, inverter, etc.

→ Distribution layer

* Distribution layer contains 4-switches.
* The 4 switches are -
(i) Main Build → R7 (Room no- 218, 2nd floor)
(ii) Main Build → R1 (Room no- 301, 3rd floor)
(iii) Main Build → R6 (Room no- 113, 1st floor)
(iv) Main Build → R2 (Room no- 201, 2nd floor)
* This 4-switches Contains -

(1) 18 port Gigabit Easy Smart Switch with 16-port. TL-SG1218MPE

(2) SG3428XMP → 24 port Gigabit & 4-port 10GESFP+, L2+ Managed switches with 24 port POE+.

→ Now let's discuss these 2 above components -

(i) TP-Link TL-SG1218MPE

* Likely used for: CCTV Camera, Wifi access point etc.
* 18 port Gigabit "Easy Smart" Switch.
* 16 port Support POE+

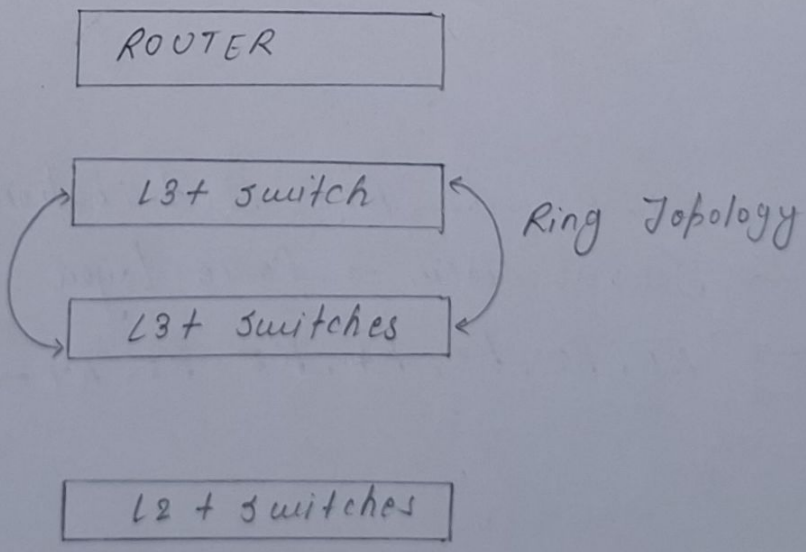
(iii) TP-Link Omada SG3428XMP

* Higher end managed layer 2+ switch.
* 24 POE+ Port, uplinks, 10G SFP
* usually acts as a main distribution for each floor.

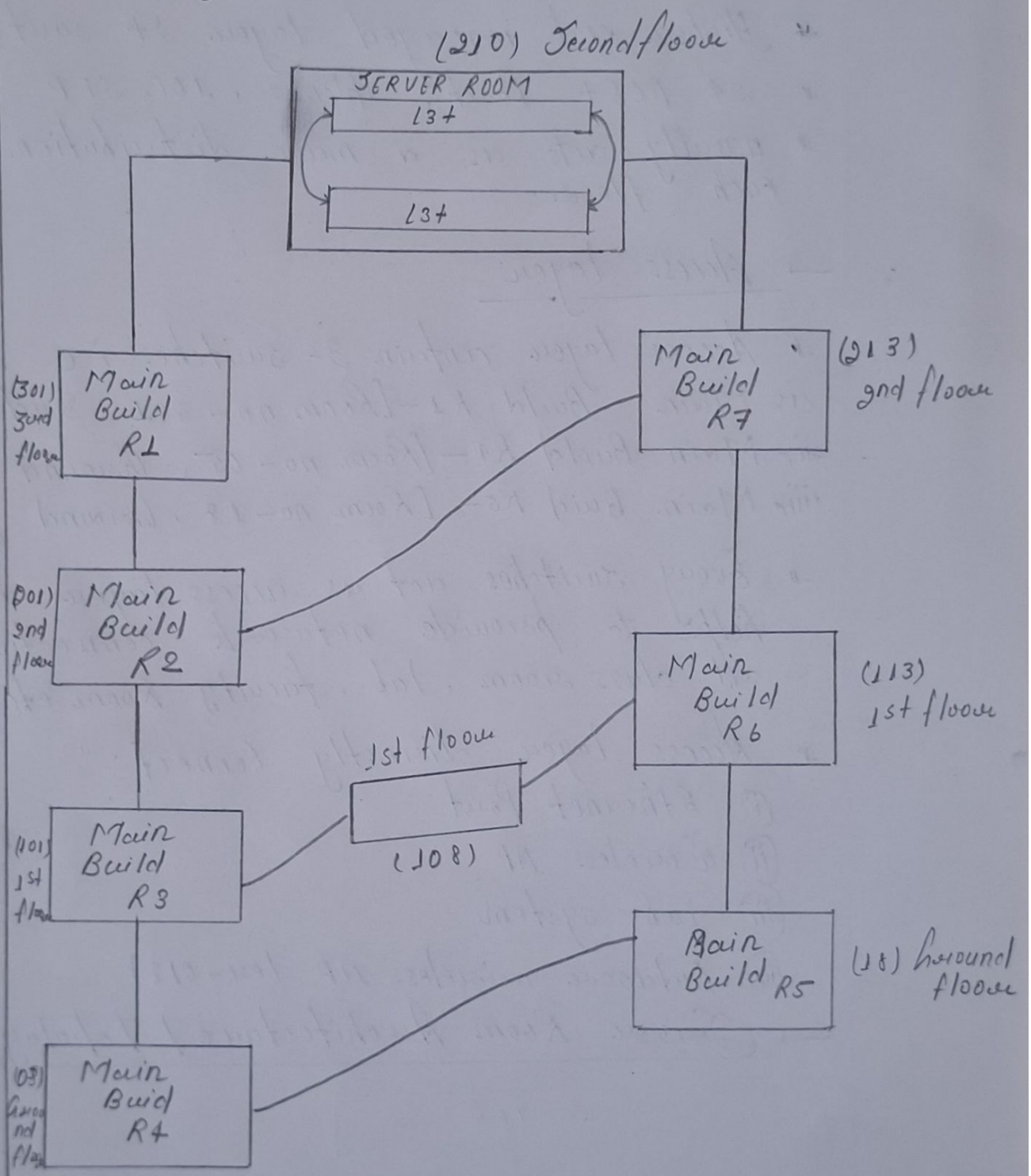
→ Access layer

* Access layer contain 3- switches i.e
(i) Main Build R1 - [Room no - 301, 3rd floor]
(ii) Main Build R4 - [Room no - 05, Ground floor]
(iii) Main Build R5 - [Room no - 18, Ground floor]
* Every switches act as access layer which helps to provide network connectively to all class room, lab, faculty Room etc.
* Access layer directly Connect:
  (i) Ethernet Port
  (ii) Wireless AP
  (iii) lab system
  (iv) Outdoor wireless AP (ex - 018)

→ Server Room Architecture / Topology



Topology

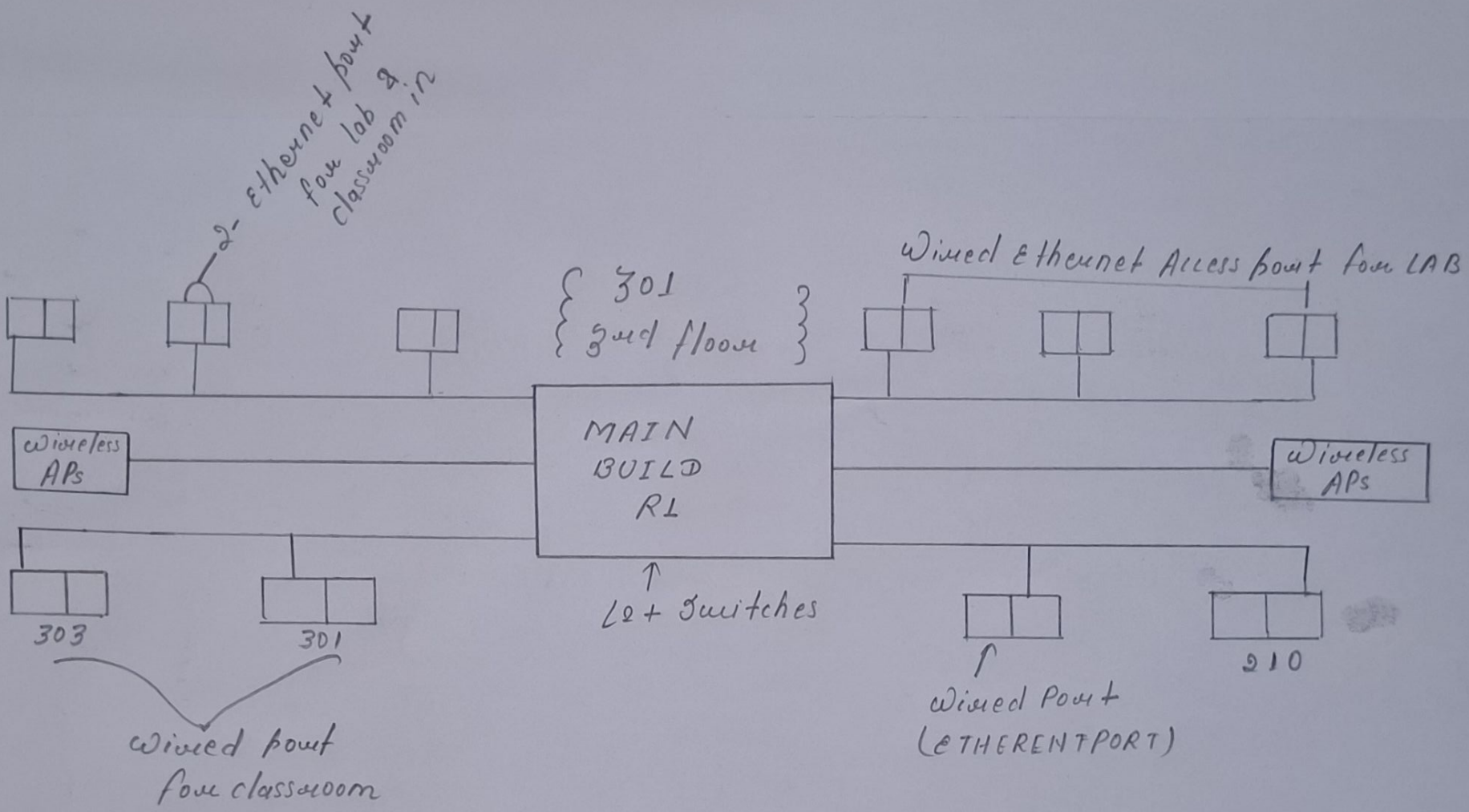


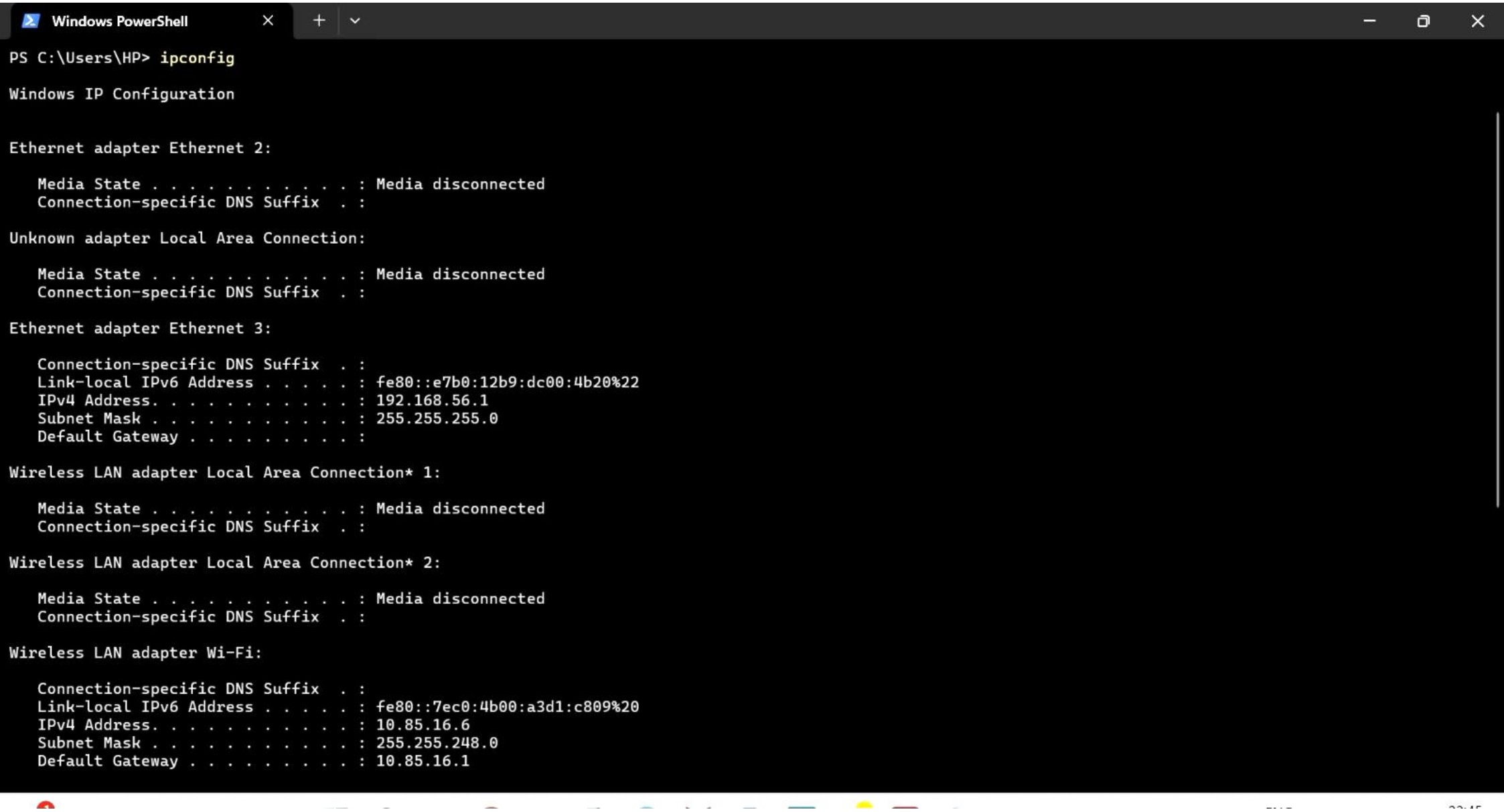
→ R1, R7, R2, R6 → Distribution layer

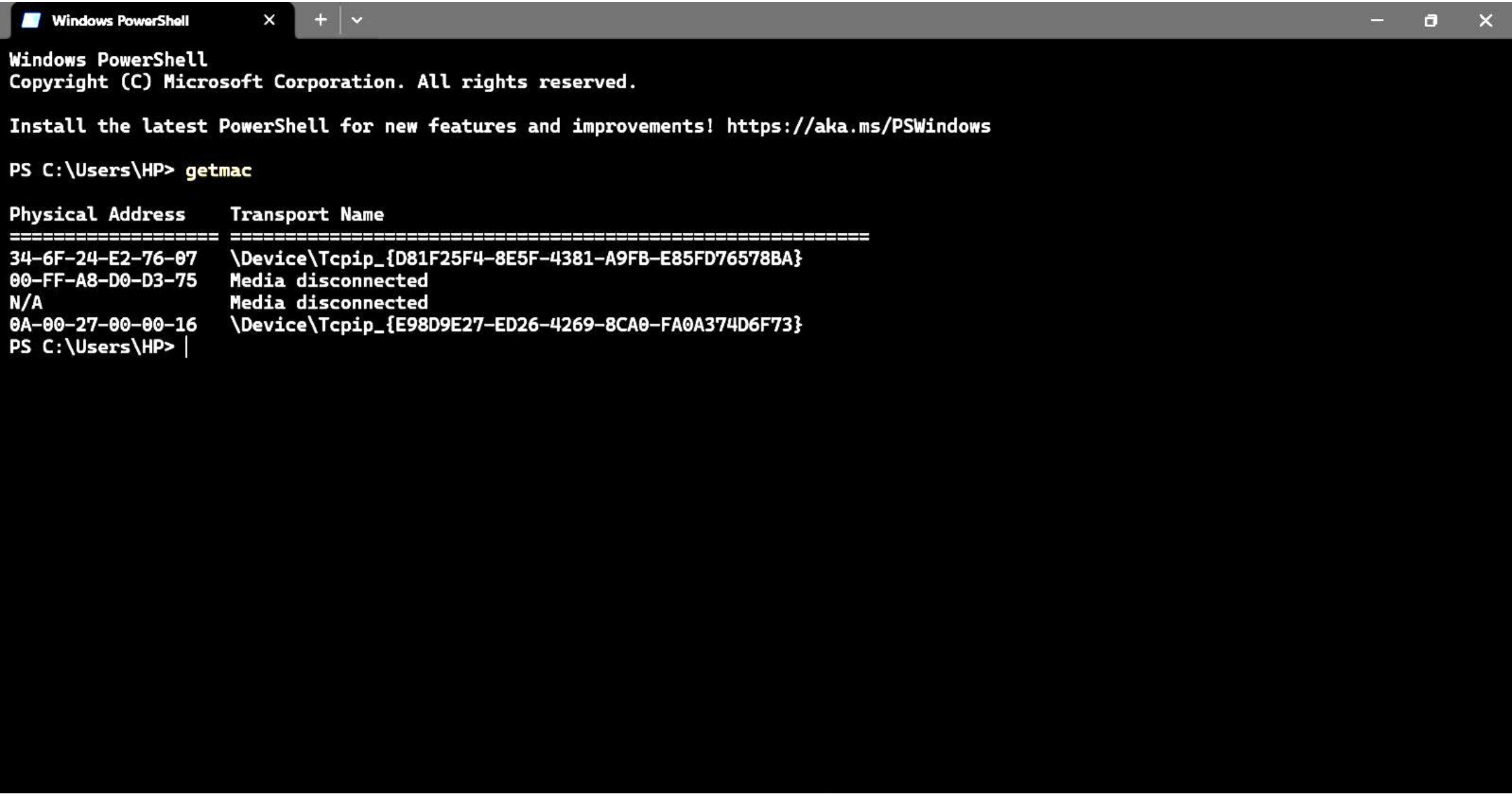
→ Server room → Core layer

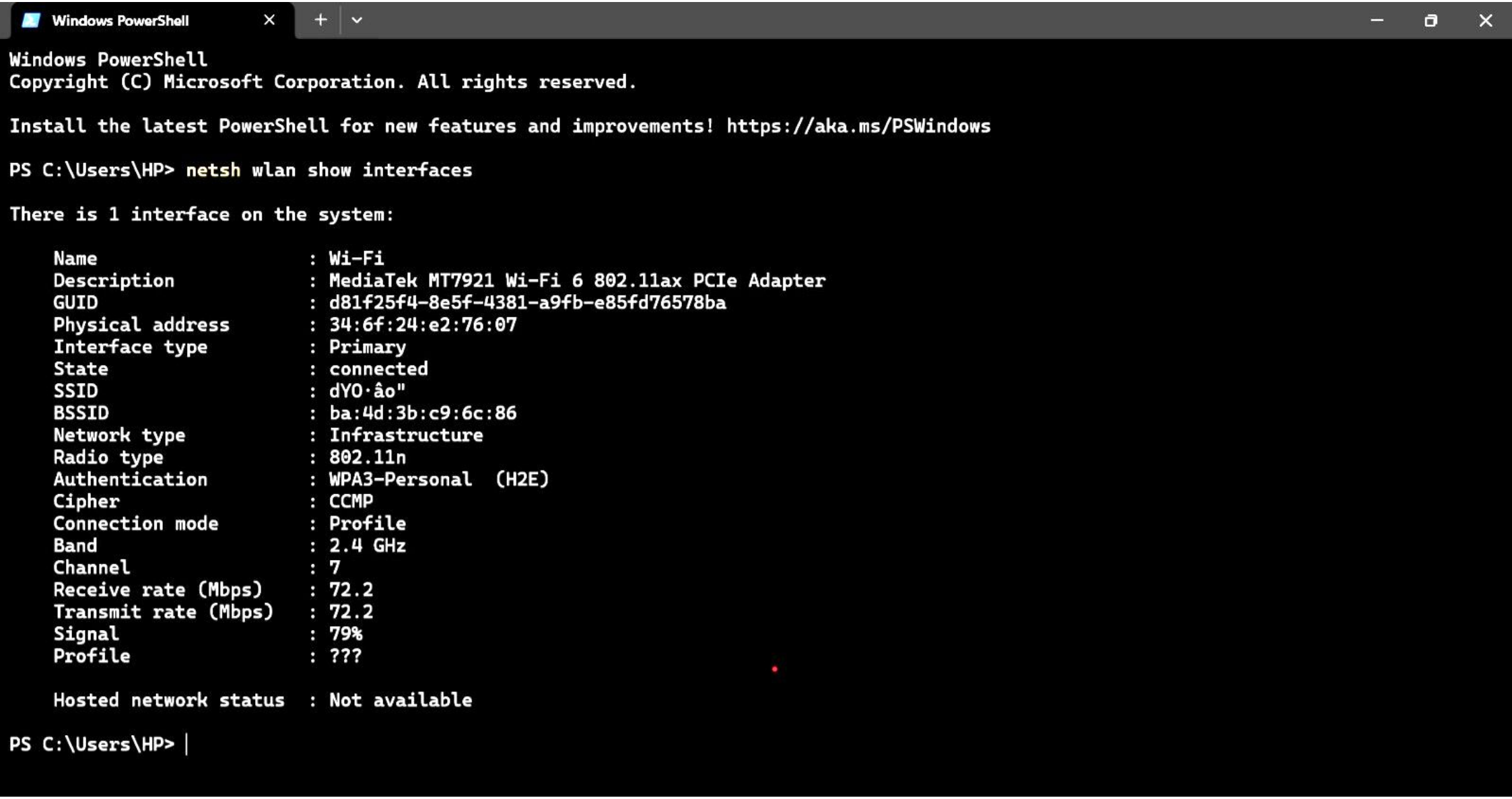
→ R1, R2, R3, R4, R5, R6, R7 → Access layer

→ Access layer Switch (3rd floor - 301)




Windows PowerShell
PS C:\Users\HP> ipconfig
Windows IP Configuration
Ethernet adapter Ethernet 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Unknown adapter Local Area Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Ethernet adapter Ethernet 3:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::e7b0:12b9:dc00:4b20%22
IPv4 Address. . . . . : 192.168.56.1
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
Wireless LAN adapter Local Area Connection* 1:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Local Area Connection* 2:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::7ec0:4b00:a3d1:c809%20
IPv4 Address. . . . . : 10.85.16.6
Subnet Mask . . . . . : 255.255.248.0
Default Gateway . . . . . : 10.85.16.1


Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\HP> getmac
Physical Address Transport Name
34-6F-24-E2-76-07 \Device\Tcpip_{D81F25F4-8E5F-4381-A9FB-E85FD76578BA}
00-FF-A8-D0-D3-75 Media disconnected
N/A Media disconnected
0A-00-27-00-00-16 \Device\Tcpip_{E98D9E27-ED26-4269-8CA0-FA0A374D6F73}
PS C:\Users\HP>


Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\HP> netsh wlan show interfaces
There is 1 interface on the system:
Name : Wi-Fi
Description : MediaTek MT7921 Wi-Fi 6 802.11ax PCIe Adapter
GUID : d81f25f4-8e5f-4381-a9fb-e85fd76578ba
Physical address : 34:6f:24:e2:76:07
Interface type : Primary
State : connected
SSID : dYO·âo"
BSSID : ba:4d:3b:c9:6c:86
Network type : Infrastructure
Radio type : 802.11n
Authentication : WPA3-Personal (H2E)
Cipher : CCMP
Connection mode : Profile
Band : 2.4 GHz
Channel : 7
Receive rate (Mbps) : 72.2
Transmit rate (Mbps) : 72.2
Signal : 79%
Profile : ???
Hosted network status : Not available
PS C:\Users\HP>

→ Difference b/w wired & wireless speed?

Ans-

Wired Network	Wireless Network
* faster communication	* slower communication
* stable connection	* Affected by distance/ Interference
* low latency	* Higher latency
* Dedicated physical medium	* Shared radio medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

→ Wireless connections uses radio wave which are affected by waves which are affected by interference, distance, obstacles & multiple connected users.

Q. Role of subnet mask in network?

Ans:- A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

* It is important for:
  (i) Network segment
  (ii) Efficient routing
  (iii) Communication b/w devices.

→ Observation —

① During the study of the college network infrastructure —

(i) A centralized Server room was used as network backbone.

(ii) Two L3 switches were connected in redundant cyclic structure.

(iii) Managed IP-link POE+ switches were used in network cabinets.

→ Advantage of Existing Network?

* Scalable architecture.
* Centrallzed management.
* Supports POE+ device.
* Easy expansion of floor & classroom
* support both wireless & wired.
* Redundant link for failover.

→ Conclusion —

The observed campus LAN follow a Hybrid hierarchical topology consists of tree, Extended star, Ring. The network uses managed POE Switches, wireless access point, ethernet port & centralized Servermfrastructure to provide efficient wired & wireless communication across the campus.